9.8 Types of Errors

Some common types of errors are given below:

Type of Error	Description
Syntax Error	• Syntax is a set of grammatical rules to construct a program. Every programming language has unique rules for constructing the sourcecode. • Syntax errors occur when grammatical rules of C++ are violated. • Example: if you type as follows, C++ will throw an error. **cout << "Welcome to Programming in C++"** • As per grammatical rules of C++, every executable statement should terminate with a semicolon. But, this statement does not end with a semicolon.
Semantic Error	• A Program has not produced expected result even though the program is grammatically correct.It may be happened by wrong use of variable / operator / order of execution etc. This means, program is grammatically correct, but it contains some logical error. So, Semantic error is also called as **"Logic Error"**.
Run-time error	• A run time error occurs during the execution of a program. It occurs because of some illegal operation that takes place. • For example, if a program tries to open a file which does not exist, it results in a run-time error

Points to Remember:

- C++ was developed by Bjarne Stroustrup at AT & T Bell Labs during the year 1979.
- Character set is the set of characters which are allowed to write C++ programs.
- Individual elements are collectively called as Lexical units or Lexical elements or Tokens.
- Keywords are the reserved words that convey specific meaning to the C++ compiler.
- Identifiers are user-defined names given to different parts of the C++ program viz. variables, functions, arrays, classes etc.,
- Literals are data items whose values do not change during the execution of a program. Therefore, Literals are called as Constants.
- There are different kinds of literals used in C++ (Integer, Float, Character, String)
- The symbols which are used to do some mathematical, logical operations are called as "Operators".
- Punctuators are symbols, which are used as delimiters in constructing C++ programs. They are also called as "Separators".
- Extraction operator(>>) and Insertion operator (<<) are used to get input and send output in C++.

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